

# FLUIDMIND SCIENTIFIC REFERENCE COMPENDIUM

OFFICIAL TEXTBOOK & LECTURE REPOSITORY • VOL. 2026

## COMPLETE WORKED SOLUTIONS & STEP-BY-STEP DERIVATIONS

### SOLUTIONS MANUAL: FLUID MECHANICS 7TH ED

Author / Compiler: Frank M. White - Prepared for Mechanical Engineering Students

Curated by: Mohammadamin Sharif (IUST Mechanical Engineering Department)

#### VOLUME OVERVIEW & ABSTRACT:

Complete official solution compendium for end-of-chapter problems in White Fluid Mechanics 7th Edition. Covers Navier-Stokes equations, Reynolds transport theorem, turbulent pipe network analysis, and drag/lift coefficient calculations.

#### SYLLABUS & CORE ENGINEERING TOPICS COVERED:

- Rigorous mathematical derivations & fundamental physical laws
- Complete worked analytical solutions with free-body & control volume diagrams
- Numerical verification & computational correlation (CFD / FEA / MATLAB)
- Practical engineering applications, safety factors, and standard design codes
- Comprehensive reference tables, property charts & dimensional constants
- Formatted for academic study, research references, and professional exam preparation

## FUNDAMENTAL EQUATIONS, WORKED METHODOLOGY & NOTES

### 1. Governing Equations & Control Volume Formulations

Conservation of mass, linear and angular momentum, and first law energy conservation applied across differential fluid elements and me

### 2. Boundary Conditions & Interface Dynamics

Evaluation of kinematic velocity continuity, no-slip wall conditions, thermal heat flux balances, and dynamic shear equilibrium across conti

### 3. Dimensionless Numbers & Similitude Criteria

Reynolds (Re), Mach (Ma), Prandtl (Pr), Nusselt (Nu), and Euler (Eu) scaling analysis ensuring experimental and numerical dynamic simi

### 4. Computational Verification & Solution Strategy

Iterative finite volume / finite element procedures, discretization error bounds, convergence criteria, and cross-comparison against empiric